

Patch-Based Diffusion Models for Solving Inverse Problems in Computed Tomography Imaging



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Abstract

This is the abstract

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1 Introduction

Computed Tomography (CT) imaging is widely used in both medicine and industry for investigating the internal structure of objects in a relatively non-destructive way. It works by directing X-rays through a sample from a number of different angles, and measuring their attenuation at each angle. Due to radiation exposure risks, industrial applications are often able to use higher energy X-rays, and take measurements at a greater number of angles (views), than is possible in medical applications. In the medical domain, however, we wish to minimise the radiation dose to the patient, resulting in fewer views being taken, and often at lower X-ray energies. This leads to artifacts from both Gaussian and Poisson noise, as well as from the limited number of views, making it difficult to accurately reconstruct the internal structure of the sample. [1].

2 Background

3 Methods

4 Results

5 Discussion

6 Conclusion

References

- [1] Ho, J., Jain, A., and Abbeel, P. (2020). Denoising Diffusion Probabilistic Models.

A LLM Usage Declaration

LLMs were used for assistance with coding, as declared in the repository README file. They were never used to generate text for inclusion in this document, and were only used to proofread and suggest minor wording alterations.

B Additional Figures

